



Barker College

**2002
TRIAL
HIGHER SCHOOL
CERTIFICATE**

Mathematics Extension 2

Staff Involved:

- DOK/GJR*
- BHC
- MRB

AM THURSDAY 8 AUGUST

40 copies

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used

Total marks (120)

- Attempt Questions 1 – 8
- All questions are of equal value
- Write your Barker Student Number on ALL pages of your answer sheets
- A table of standard integrals is provided on page 10
- ALL necessary working should be shown in every question
- Start each question on a NEW page
- Write on one side of each answer page
- Marks may be deducted for careless or badly arranged work

Total marks (120)

Attempt Questions 1 – 8

ALL questions are of equal value

Answer each question on a SEPARATE sheet of paper

Marks

Question 1 (15 marks) [BEGIN A NEW PAGE]

- (a) Find $\int_1^e \frac{dx}{x(1 + (\ln x)^2)}$ by substituting $u = \ln x$ 2
- (b) Find $\int \frac{x + 1}{x^2 + 4} dx$ 2
- (c) Find $\int \frac{x^2 + 4}{x + 1} dx$ 3
- (d) Evaluate $\int_0^{\frac{\pi}{4}} x^2 \sin x dx$ 4
- (e) Prove that $\int_0^{\frac{1}{4}} \sqrt{1 - 4x^2} dx = \frac{\pi}{24} + \frac{\sqrt{3}}{16}$ 4

Marks

Question 2 (15 marks) [BEGIN A NEW PAGE]

(a) Given that $f(x) = e^{-x}$, sketch the following showing the main features.

(i) $y = -f(x)$ **1**

(ii) $y = 1 - f(x)$ **2**

(iii) $y = \frac{1}{1 - f(x)}$ **2**

(iv) $y = \left| \frac{1}{1 - f(x)} \right|$ **2**

(b) Next to each graph state whether it is odd, even or neither. **4**

(c) (i) For $x^2 + 2xy + y^5 = 4$, show that $\frac{dy}{dx} = \frac{-2x - 2y}{2x + 5y^4}$ **2**

(ii) A plane curve is defined implicitly by the equation

$$x^2 + 2xy + y^5 = 4.$$

This curve has a horizontal tangent at the point $P(x_1, y_1)$.

Show that x_1 is a root of the equation $x^5 + x^2 + 4 = 0$. **2**

Marks

Question 3 (15 marks) [BEGIN A NEW PAGE]

(a) If $z_1 = 1 + 2i$, $z_2 = 2 - i$ and $z_3 = -1 + i\sqrt{3}$, find $\left| \frac{z_1 z_2}{iz_3} \right|$ 3

(b) Simplify $\frac{(2\cos\theta + 2i\sin\theta)^5 (2\cos\theta + 2i\sin\theta)^{-3}}{(\cos 2\theta + i\sin 2\theta)^5}$ 3

(c) Z is the point representing the complex number z on an Argand diagram.

(i) Describe in words the geometrical significance of the expressions

$$|z - 2| \quad \text{and} \quad \operatorname{Re}(z)$$
2

(ii) Hence, or otherwise, sketch the locus of Z given that

$$|z - 2| = \operatorname{Re}(z)$$

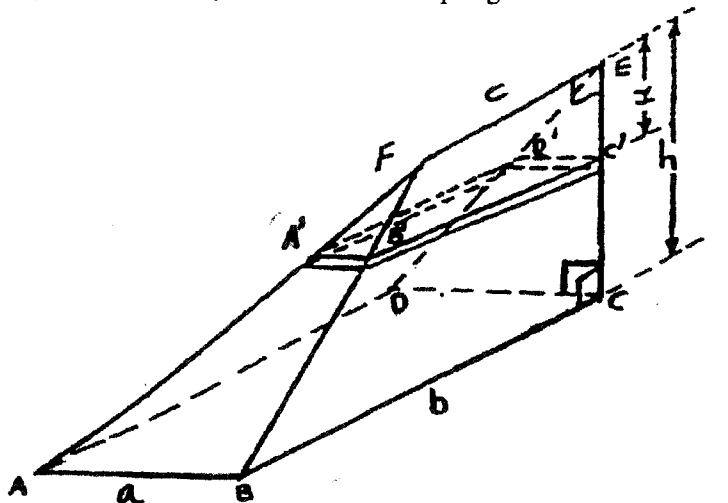
Show all important features of this locus. 3

(d) Triangle OAB is an isosceles triangle with $AO = OB$ and $\angle OBA = 75^\circ$.
If O is the origin and A represents the complex number $-\sqrt{3} + i$, find **two**
possible complex numbers represented by the point B , in the form $a + bi$. 4

Question 4 (15 marks) [BEGIN A NEW PAGE]

Marks

- (a) Consider solid ABCDEF whose height is h , and whose base is a rectangle ABCD, where $AB = a$, $BC = b$ and the top edge $EF = c$.



Consider a rectangular slice $A'B'C'D'$ (parallel to the base ABCD) which is x units from the top edge with width Δx .

NOTE: $B'C' \parallel BC$ and $A'B' \parallel AB$

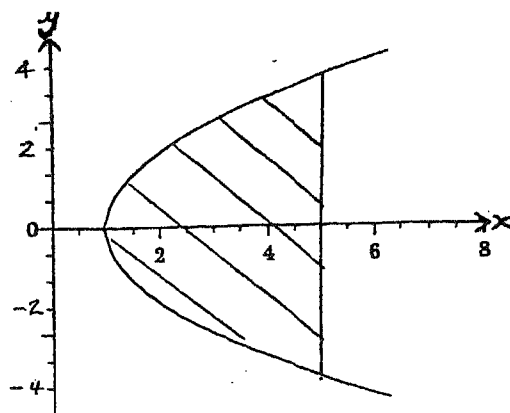
- (i) Show that the volume Δv of the slice is given by

$$\Delta v = \left(\frac{x}{h} a \right) \left(c + \frac{b-c}{h} x \right) \Delta x \quad 4$$

- (ii) Hence, show that the volume of the solid ABCDEF is

$$\frac{ha}{6} (2b + c) \quad 4$$

- (b) The diagram shows the region bounded by the curve $y^2 = 4(x - 1)$ and the line $x = 5$. By using the method of cylindrical shells, or otherwise, find the volume of the solid formed by rotating the given region about the y -axis. 7



Marks

Question 5 (15 marks) [BEGIN A NEW PAGE]

The normal at $P(ct, \frac{c}{t})$ to the hyperbola $xy = c^2$ meets the curve again at Q .

- (a) Prove that the equation of the normal is $t^3x - ty = ct^4 - c$ 4
- (b) Find the coordinates of Q . 3
- (c) A line from P through the origin meets the hyperbola again at R .
Prove that PR is perpendicular to QR . 4
- (d) If M is the midpoint of PQ , find the equation of the locus M . 4

Marks

Question 6 (15 marks) [BEGIN A NEW PAGE]

(a) α and β are the complex roots of $iz^2 + \sqrt{3}z - 1 = 0$.

(i) Find α and β in $a + ib$ form. **3**

(ii) Show that $\alpha^2\beta^2 + 1 = 0$. **1**

(b) Solve the equation $4x^3 - 12x^2 + 11x - 3 = 0$ given that the roots are in arithmetic sequence. **4**

(c) (i) Prove, by calculus if you wish, that the polynomial equation

$$\frac{1}{4}x^4 - \frac{1}{3}x^3 - 2x^2 + 4x + c = 0$$

has no real roots if $c > 9\frac{1}{3}$ **5**

(ii) Find an approximation for the **largest** root of the polynomial equation in (i) above, if $c = -2$, using one application of Newton's Method. **2**

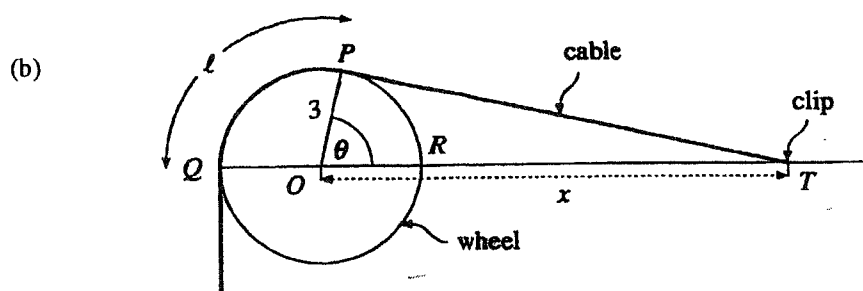
Question 7 (15 marks) **[BEGIN A NEW PAGE]**

Marks

(a) Let n be a positive integer where $I_n = \int_1^2 (\log_e x)^n dx$

(i) Prove that $I_n = 2(\log_e 2)^n - nI_{n-1}$ 3

(ii) Hence, evaluate $\int_1^2 (\log_e x)^4 dx$ 2



A long cable is wrapped over a wheel of radius 3 metres and one end is attached to a clip at T . The centre of the wheel is at O and QR is a diameter. The point T lies on the line OR at a distance x metres from O .

The cable is tangential to the wheel at P and Q as shown.

Let $\angle POR = \theta$ (in radians).

The length of cable in contact with the wheel is ℓ metres; that is, the length of the arc between P and Q is ℓ metres.

(i) Explain why $\cos \theta = \frac{3}{x}$ 1

(ii) Show that $\ell = 3 \left[\pi - \cos^{-1} \left(\frac{3}{x} \right) \right]$ 2

(iii) Show that $\frac{d\ell}{dx} = \frac{-9}{x\sqrt{x^2 - 9}}$ 2

(iv) What is the significance of the fact that $\frac{d\ell}{dx}$ is negative? 1

(v) Let $s = \ell + PT$
Given that $PT^2 = QT \times RT$, or otherwise, express s in terms of x 1

(vi) The clip at T is moved away from O along the line OR at a constant speed of 2 metres per second.
Find the rate at which s changes when $x = 10$ 3

Question 8 (15 marks) [BEGIN A NEW PAGE]

- (a) It is given that the equation $ax^4 + 4bx + c = 0$ has a double root.
If α is the double root, show that $a\alpha^3 + b = 0$ and deduce that $ac^3 = 27b^4$

4

- (b) $P(x)$ is divided by $(x - a)(x - b)$ so that a remainder $R(x)$ is obtained.
Show that the remainder is given by

$$R(x) = \left(\frac{P(a) - P(b)}{a - b} \right)x + \frac{aP(b) - bP(a)}{a - b}$$

4

- (c) Using the fact that $\cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$, or otherwise,

- (i) find a general solution of the equation $\sin 3x = -\cos 2x$

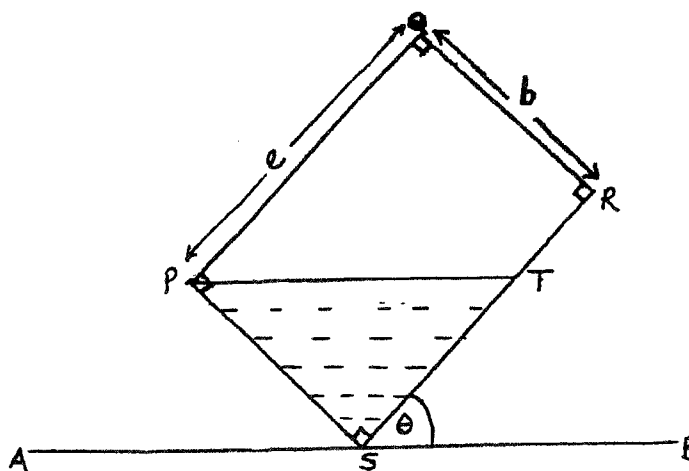
3

- (ii) find the smallest positive solution of the equation

$$\sin 3x = -\cos 2x$$

1

- (d) A rectangular fish tank PQRS is tilted at an angle of θ to the horizontal surface AB.
The surface of the water is PT, $QR = b$ and $RS = e$.



If the fish tank is lowered so that SR lies on AB, prove that the height, h , of the water in the tank is given by

$$h = \frac{b^2 \cot \theta}{2e}$$

3

End of Paper

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RIAL SOLUTIONS

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Question One

$$(a) \int_1^e \frac{dx}{x(1+(\ln x)^2)} = \int_0^1 \frac{du}{1+u^2} \quad (1)$$

$$\left. \begin{aligned} \text{Let } u &= \ln x \\ du &= \frac{1}{x} dx \\ &= \tan^{-1} u \Big|_0^1 \\ &= \tan^{-1} 1 - \tan^{-1} 0 \\ &= \pi/4 \end{aligned} \right\} (1)$$

$$(b) \int \frac{x+1}{x^2+4} dx = \frac{1}{2} \int \frac{2x}{x^2+4} dx + \int \frac{1}{x^2+4} dx \quad (1)$$

$$= \frac{1}{2} \ln(x^2+4) + \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C \quad (1)$$

$$(c) \int \frac{x^2+4}{x+1} dx = \int (x-1) + \frac{5}{x+1} dx \quad (1)$$

$$= \frac{x^2}{2} - x + 5 \ln(x+1) + C \quad (1)$$

$$(d) \int_0^{\pi/4} x^2 \sin x dx = -x^2 \cos x \Big|_0^{\pi/4} + 2 \int_0^{\pi/4} x \cos x dx \quad \text{by parts with } u=x^2, dv=\sin x dx$$

$$= -\frac{\pi^2}{16\sqrt{2}} + 2 \int_0^{\pi/4} x \cos x dx$$

$$= -\frac{\pi^2}{16\sqrt{2}} + 2x \sin x \Big|_0^{\pi/4} - 2 \int_0^{\pi/4} \sin x dx \quad \text{by parts with } u=x, dv=\cos x dx$$

$$= -\frac{\pi^2}{16\sqrt{2}} + 2x \sin x \Big|_0^{\pi/4} + 2 \cos x \Big|_0^{\pi/4} \quad (1)$$

- 2 -

Q1 cos

$$= \frac{-\pi^2}{16\sqrt{2}} + \frac{\pi}{2\sqrt{2}} + 2 \left[\cos x \right]_0^{\pi/4} \quad (1)$$

$$= \frac{-\pi^2}{16\sqrt{2}} + \frac{\pi}{2\sqrt{2}} + \sqrt{2} - 2 \quad (1)$$

$$= \sqrt{2} - 2 + \frac{8\pi - \pi^2}{16\sqrt{2}} \quad 4$$

$$(e) \int_0^{\pi/4} \sqrt{1-4x^2} dx = 2 \int_0^{\pi/4} \sqrt{\frac{1}{4} - x^2} dx \quad (1)$$

$$y = \sqrt{\frac{1}{4} - x^2} \quad \left(\frac{1}{4}, \frac{\sqrt{3}}{4}\right)$$

$$= 2 \times \text{Area}_{\Delta} + 2 \times \text{Area}_{\text{sector}} \quad (1)$$

$$= 2 \times \left(\frac{1}{2} \times \frac{1}{4} \times \frac{\sqrt{3}}{4} \right) + 2 \times \left(\frac{1}{12} \times \pi \times \left(\frac{1}{2} \right)^2 \right)$$

$$= \frac{\sqrt{3}}{16} + \frac{\pi}{24} \quad (1)$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{3}/4}{1/4}\right) = \tan^{-1}\sqrt{3} = 60^\circ$$

$$\therefore \phi = 30^\circ = \frac{1}{12} \text{ of circle area}$$

Alt: use trig substitution

$$x = \frac{1}{2} \sin \theta$$

$$x^2 = \frac{1}{4} \sin^2 \theta$$

$$\sqrt{1-4x^2} = \cos \theta$$

$$dx = \frac{1}{2} \cos \theta d\theta$$

$$x=0 \Rightarrow \sin \theta = 0 \Rightarrow \theta = 0$$

$$x = \frac{1}{4} \Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$$

$$\int_0^{\pi/4} \sqrt{1-4x^2} dx = \int_0^{\pi/6} \frac{1}{2} \cos^2 \theta d\theta \quad (1) \text{ includes substitution}$$

$$= \frac{1}{4} \int_0^{\pi/6} (1 + \cos 2\theta) d\theta \quad (1)$$

$$= \frac{1}{4} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi/6} \quad (1)$$

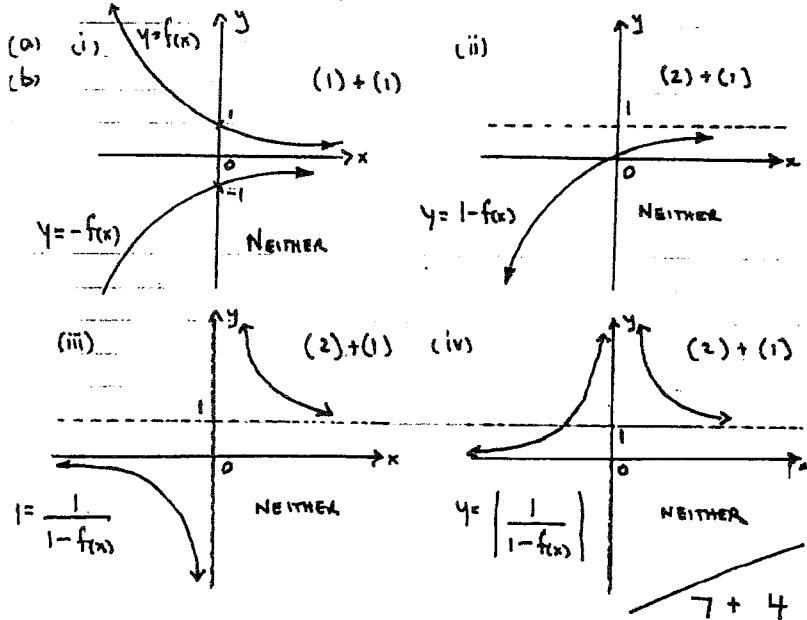
$$= \frac{1}{4} \left(\frac{\pi}{6} + \frac{\sqrt{3}}{2} \right)$$

$$= \frac{\sqrt{3}}{8} + \frac{\pi}{24} \quad (1)$$

- 3 -

Question Two

$$y = f(x) = e^{-x}$$



(i) (i) $x^2 + 2xy + y^5 = 4$

$$\Rightarrow 2x + (2xy' + 2y) + 5y^4 y' = 0 \quad (1)$$

$$\Rightarrow y'(2x + 5y^4) = -2x - 2y$$

$$\Rightarrow y' = \frac{-2x - 2y}{2x + 5y^4} \quad (1)$$

2

- 4 -

Q2 cont (c) (ii) $\frac{dy}{dx} = 0 \Rightarrow -2x - 2y = 0$
 $\Rightarrow y = -x$

so $P(x, y) = P(x, -x) \quad (1)$

put P in $x^2 + 2xy + y^5 = 4$
 $\Rightarrow x_1^2 + 2x_1(-x_1) + (-x_1)^5 = 4$
 $\Rightarrow -x_1^2 - x_1^5 = 4$
 $\Rightarrow x_1^5 + x_1^2 + 4 = 0$

hence x_1 is a root of $x^5 + x^2 + 4 = 0$

2

Question Three

a. $Z_1 = 1 + 2i \Rightarrow |Z_1| = \sqrt{5}$
 $Z_2 = 2 - i \Rightarrow |Z_2| = \sqrt{5} \quad (1)$
 $Z_3 = -1 + i\sqrt{5} \Rightarrow |Z_3| = 2$

$$\frac{|Z_1 Z_2|}{|i Z_3|} = \frac{|Z_1| |Z_2|}{|i| |Z_3|} \quad (1)$$

$$= \frac{\sqrt{5} \times \sqrt{5}}{1 \times 2} = \frac{5}{2}$$

3

b. $\frac{(2\cos\theta + i\sin\theta)^5 (2\cos\theta + i\sin\theta)^{-3}}{(2\cos\theta + i\sin\theta)^5} = \frac{(2\cos\theta + i\sin\theta)^{-8}}{(2\cos\theta + i\sin\theta)^5} \quad (1)$

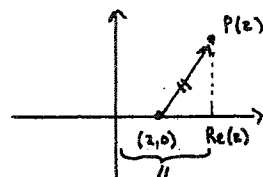
$$= \frac{4}{(2\cos\theta + i\sin\theta)^3} \text{ or } 4(\cos\theta + i\sin\theta)^{-3} \quad (1)$$

3

c) (i) $|Z - 2|$ is the vector from (2, 0) to P(z) (1)

Re(z) is the x-axis position of P(z) (1)
 (vector from (0, 0) to)

Q3 (c) (ii)

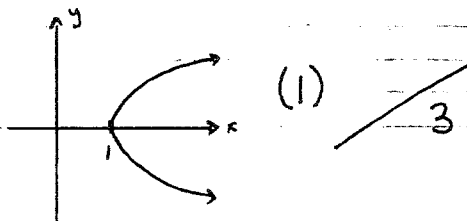


$$|z - z| = \text{Re}(z) \quad \text{let } z = x + iy$$

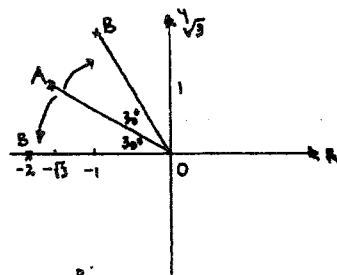
$$\Rightarrow |(x + iy) - z| = \text{Re}(x + iy) \quad (1)$$

$$\begin{aligned} \Rightarrow \sqrt{(x - z)^2 + y^2} &= x \\ (x - z)^2 + y^2 &= x^2 \\ x^2 - 4x + 4 + y^2 &= x^2 \\ \Rightarrow y^2 &= 4x - 4 = 4(x - 1) \end{aligned} \quad (1)$$

Hence Locus is:



(d)



$$\begin{aligned} |OA| &= 2 \\ \arg(OA) &= \pi - \tan^{-1}\left(\frac{1}{2}\right) = \frac{5\pi}{6} \end{aligned}$$

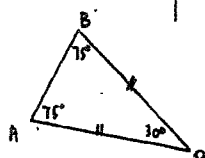
$$\therefore \angle AOB = 30^\circ$$

If $\triangle OAB$ isos and $\angle OBA = 75^\circ$ as $\angle AOB = 30^\circ$

Want two vectors length 2, 30° either side of $\vec{OA}$

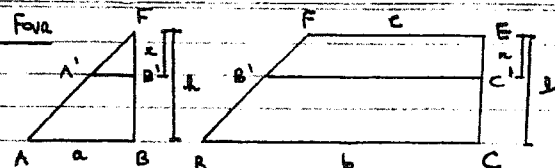
$$\begin{aligned} \text{From diagram } \vec{OB} &= -2 \\ \text{or } \vec{OB} &= 2 \text{cis } \frac{2\pi}{3} \end{aligned} \quad (1)$$

$$= -1 + i\sqrt{3} \quad (1)$$



Question Four

(a)



(1) For setup of Problem

$$\begin{aligned} \text{at } x=0 \quad z=0 \\ x=h \quad z=a \end{aligned}$$

$$\begin{aligned} \text{at } x=0 \quad y=c \\ x=h \quad y=b \end{aligned}$$

$$z = mx + b$$

$$0 = b$$

$$a = mh \Rightarrow m = \frac{a}{h}$$

$$\therefore z = \left(\frac{a}{h}\right)x \quad (1)$$

$$y = mx + d$$

$$c = d$$

$$b = mh + d$$

$$mh = b - c$$

$$\Rightarrow m = \frac{b - c}{h}$$

$$\therefore y = \left(\frac{b - c}{h}\right)x + c \quad (1)$$

at height x

$$A(x) = zy$$

$$= \left(\frac{xa}{h}\right) \left(c + \left(\frac{b - c}{h}\right)x\right)$$

for a slice of Δx chosen sufficiently small

$$\Rightarrow \Delta V = A(x) \Delta x \quad (1)$$

$$= \left(\frac{xa}{h}\right) \left(c + \left(\frac{b - c}{h}\right)x\right) \Delta x$$

Q4 (a) (ii)

from (i) $\Rightarrow V = \int_0^h \frac{\Delta V}{\Delta x} dx$

$$= \int_0^h \left(\frac{xh}{h} \right) \left(c + \frac{b-c}{h} x \right) dx \quad (1)$$

$$= \frac{a}{h} \int_0^h \left(xc + \frac{b-c}{h} x^2 \right) dx$$

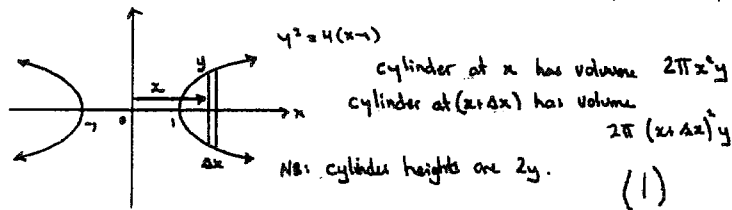
$$= \frac{a}{h} \left[\frac{xc^2}{2} + \frac{1}{3} \left(\frac{b-c}{h} \right) x^3 \right]_0^h \quad (1)$$

$$= \frac{a}{h} \left[\frac{ch^2}{2} + \frac{1}{3} \left(\frac{b-c}{h} \right) h^3 \right] \quad (1)$$

$$= \frac{ach}{2} + \frac{ah(b-c)}{3}$$

$$= \frac{ah}{6} \{ 2b + c \} \quad (1)$$

(b)



$\therefore$ Volume of shell is $2\pi (x+\Delta x)^2 y - 2\pi x^2 y \quad (1)$

$$= 2\pi y (x^2 + 2x\Delta x + \Delta x^2 - x^2)$$

$$= 2\pi y (2x\Delta x + \Delta x^2) \quad (1)$$

as Δx is sufficiently small, ignore $\Delta x^2 \Rightarrow \Delta V = 4\pi xy \Delta x \quad (1)$

Q4 (b) cont

$\therefore V = \int_1^5 \frac{\Delta V}{\Delta x} dx$

$$= \int_1^5 4\pi xy \, dx$$

$(**) = 4\pi \int_1^5 x \cdot 2\sqrt{x-1} \, dx \quad \text{as } y = \pm 2\sqrt{x-1}$

$$= 8\pi \int_1^5 x\sqrt{x-1} \, dx \quad (1)$$

let $u = x-1$
 $x = u+1$
 $du = dx$

$$= 8\pi \int_0^4 (u+1)\sqrt{u} \, du$$

$$= 8\pi \left[\frac{2u^{5/2}}{5} + \frac{2u^{3/2}}{3} \right]_0^4 \quad (1)$$

$$= 8\pi \left[\frac{64}{5} + \frac{16}{3} \right]$$

$$= \frac{2176\pi}{15} \text{ Units}^3 \quad (1)$$

Ans at $(*)$ use

$$u^2 = x-1$$

$$x = u^2 + 1$$

$$dx = 2u \, du$$

at $(**)$

substitute to y instead of x

$$y^2 = 4(x-1) \Rightarrow x = \frac{y^2}{4} + 1$$

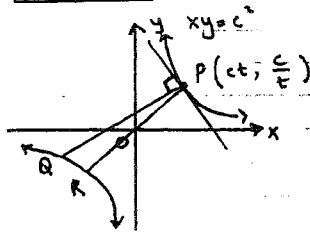
$$dx = \frac{y}{2} dy$$

end up with $8\pi \int_0^2 (2u^2 + 2u) du$

end up with $4\pi \int_0^4 \left(\frac{y^2}{4} + 1 \right) \cdot y \cdot \frac{y}{2} dy$

$$= \frac{\pi}{2} \int_0^4 (y^4 + 4y^2) dy$$

Question Five



(a) $xy = c^2 \Rightarrow xy' + y = 0 \Rightarrow y' = -\frac{y}{x}$ (1)

at P $\Rightarrow y' = \frac{-y}{x} = \frac{-c/t}{ct} = -\frac{1}{t^2}$
 $\therefore$ grad of normal is t^2 (1)

eqn of normal: $y - \frac{c}{t} = t^2(x - ct)$ (1)

$\Rightarrow y - \frac{c}{t} = t^2x - ct^3$

$\Rightarrow ty - c = t^3x - ct^4$ (1)

or $t^3x - ty = ct^4 - c$ as required. 4

(ii) Solve $t^3x - ty = ct^4 - c$ — (1)
 $xy = c^2$ — (2)

put (2) in (1) $\Rightarrow t^3x - \frac{c^2}{x} = ct^4 - c$ (1)
 $y = c^2/x$

$\therefore t^3x^2 - c(t^4 - 1)x - c^2t = 0$

$\therefore x = \frac{c(t^4 - 1) \pm \sqrt{c^2(t^4 - 1)^2 + 4t^3c^2}}{2t^3}$
 $= \frac{c(t^4 - 1) \pm \sqrt{c^2(t^4 + 1)^2}}{2t^3}$ (1)

$= \frac{2ct^4}{2t^3}$ or $\frac{-2c}{2t^3}$

$= ct$ or $-\frac{c}{t^3}$ (1) 3

Hence $y = \frac{c}{t}$ or $-ct^3 \therefore Q = (-\frac{c}{t^3}, -ct^3)$

(c) by symmetry $R = (-ct, -\frac{c}{t})$ (1)

$G_{PR} = \frac{\frac{c}{t} - \frac{c}{t}}{ct + ct} = \frac{\frac{0}{t}}{2ct} = \frac{1}{t^2}$ (1)

$G_{QR} = \frac{-ct^3 + \frac{c}{t}}{-\frac{c}{t^3} + ct} = \frac{-ct^4 + ct^2}{-c + ct^4} = \frac{-ct^2(1 - t^2)}{-c(1 - t^4)} = -t^2$ (1)

$G_{PR} \cdot G_{QR} = \frac{1}{t^2} \times -t^2 = -1$ Hence $PR \perp QR$ (1) 4

(d) $M = \left(\frac{ct - \frac{c}{t^3}}{2}, \frac{\frac{c}{t} - ct^3}{2} \right)$

Locus of M:

Let $x = \frac{ct - \frac{c}{t^3}}{2}$ and $y = \frac{\frac{c}{t} - ct^3}{2}$
 $= \frac{c(t^4 - 1)}{2t^3} = \frac{c(1 - t^4)}{2t}$

Eliminate t :

$\frac{2x}{c} = \frac{t^4 - 1}{t^3}$ — (1) $-\frac{2y}{c} = \frac{t^4 - 1}{t}$ — (2)

Now (2) $\div$ (1) $\Rightarrow \frac{-y}{x} = t^2$ — (3)

(2) cubed $\Rightarrow \left(\frac{-2y}{c} \right)^3 = \left(\frac{t^4 - 1}{t} \right)^3 = \frac{(t^4 - 1)^3}{t^3}$
 $= (t^4 - 1)^2 \cdot \left(\frac{t^4 - 1}{t^3} \right)$

with (1) $= (t^4 - 1)^2 \cdot \left(\frac{2x}{c} \right)$ (1)

$$\left(\frac{c}{2x}\right) \times \frac{-8y^3}{c^3} = (t^4 - 1)^2$$

$$\text{So } t^4 - 1 = \sqrt{\frac{-4y^3}{c^3 x}}$$

$$\therefore t^4 = 1 + \frac{2y}{c} \sqrt{\frac{-y}{x}} \quad \text{--- (4)}$$

$$\text{Sub (4)} \Rightarrow t^4 = \frac{y^2}{x^2} = 1 + \frac{2y}{c} \sqrt{\frac{-y}{x}} \quad \text{--- (1)}$$

$$\Rightarrow \frac{y^2}{x^2} - 1 = \frac{2y}{c} \sqrt{\frac{-y}{x}} \quad \text{--- (1)}$$

$$\Rightarrow \left(\frac{cy}{2x^2} - \frac{c}{2y}\right)^2 = \frac{-y}{x}$$

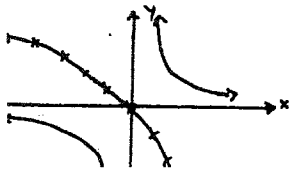
$$\Rightarrow \frac{c^2 y^2}{4x^4} + \frac{c^2}{4y^2} - \frac{2c^2 y}{2x^2 y} = \frac{-y}{x}$$

$$\Rightarrow c^2 \left(\frac{y^2}{x^4} + \frac{1}{y^2} - \frac{2}{x^2} \right) = \frac{-4y}{x}$$

$$\times x^4 y^2 \Rightarrow c^2 (y^4 + x^4 - 2x^2 y^2) = -4x^3 y^3$$

$$\Rightarrow c^2 (x^2 - y^2)^2 = -4x^3 y^3$$

$$\therefore c^2 (y^2 - x^2)^2 + 4x^3 y^3 = 0 \text{ is the locus (1)}$$



Geometrically it behaves like half a hyperbola but is in fact not hyperbolic.

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Question 5m

(a) (i) $iZ^2 + \sqrt{3}Z - 1 = 0$

$$\begin{aligned} \text{As } iZ^2 + \sqrt{3}Z - 1 &= 0 \\ \text{Let } Z &= x + iy \\ \therefore i(x + iy)^2 + \sqrt{3}(x + iy) - 1 &= 0 \\ i(x^2 - y^2 + 2ixy) + \sqrt{3}x + i\sqrt{3}y - 1 &= 0 \\ i(x^2 - y^2) - 2xy + \sqrt{3}x + i\sqrt{3}y - 1 &= 0 \\ \therefore \begin{cases} x^2 - y^2 - 2xy = \frac{1 - \sqrt{3}x}{i} \\ \sqrt{3}y = \frac{1 - \sqrt{3}x}{i} \end{cases} \end{aligned}$$

$$\begin{aligned} \text{(ii) } \alpha^2 \beta^2 + 1 &= (\alpha\beta)^2 + 1 \text{ now } \alpha\beta = \prod \text{ of roots} \\ &= (i)^2 + 1 = -1 + 1 = 0 \end{aligned}$$

b) $4x^3 - 12x^2 + 11x - 3 = 0$ --- (1)

Possible 3 roots in A.P.

thus, let roots be $a-d, a, a+d$ --- (1)

$$\sum \alpha = 3a$$

$$\sum \alpha\beta = a(a-d) + a(a+d) + (a-d)(a+d) = 3a^2 - d^2$$

$$\sum \alpha\beta\gamma = a(a-d)(a+d) = a^3 - ad^2 \quad \text{--- (1)}$$

$$\text{Also from (1) } \begin{cases} \sum \alpha = 3 \\ \sum \alpha\beta = 1/4 \\ \sum \alpha\beta\gamma = 3/4 \end{cases} \begin{cases} \text{equating} \\ \text{with} \\ \text{above} \end{cases} \Rightarrow \begin{cases} 3a = 3 \Rightarrow a = 1 \\ 3a^2 - d^2 = 1/4 \Rightarrow 3 - d^2 = 1/4 \Rightarrow d^2 = 11/4 \Rightarrow d = \sqrt{11}/2 \\ a^3 - ad^2 = 3/4 \Rightarrow 1 - d^2 = 3/4 \Rightarrow d^2 = 1/4 \Rightarrow d = 1/2 \end{cases}$$

Q6 (c) cont

(i) As it is a quartic  shape will only have no real roots if minimum values > 0

$$\text{Let } y = \frac{1}{4}x^4 - \frac{1}{3}x^3 - 2x^2 + 4x + c = 0$$

$$y' = x^3 - x^2 - 4x + 4$$

$$y'' = 3x^2 - 2x - 4$$

$$y' = 0 \Rightarrow x^3 - x^2 - 4x + 4 = 0 \quad (1)$$

$$(x-1)(x-2)(x+2) = 0$$

$\therefore x = -2, 1, 2$ are turning points

easily seen that $x = -2$ min
 $\begin{matrix} 1 & \text{max} \\ 2 & \text{min} \end{matrix} \quad (1)$

$$\text{at } x = -2 \Rightarrow y = 4 + \frac{8}{3} - 8 - 8 + c = -9\frac{1}{3} + c$$

$$\text{for no roots } y > 0 \Rightarrow c > 9\frac{1}{3}$$

$$\text{at } x = 2 \Rightarrow y = \frac{4}{3} + c$$

$$\text{for no roots } y > 0 \Rightarrow c > -\frac{4}{3}$$

as this is smaller than $9\frac{1}{3}$

need $c > 9\frac{1}{3}$ to ensure both minimums above x-axis.

(1)

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Q6 (c) (iii) cont

$$(ii) \text{ Let } y = \frac{1}{4}x^4 - \frac{1}{3}x^3 - 2x^2 + 4x - 2$$

$\Rightarrow$ largest root is when $x > 2$ (i.e. $x=2$ a min)

$$y' = x^3 - x^2 - 4x \quad (1)$$

$$\text{Let } x_0 = 3 \Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad \text{Newton's Method}$$

$$= 3 - \frac{18/4}{10}$$

$$= \frac{107}{40} \quad (1)$$

$$\approx 2.675$$

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Question Seven

$$(a) \text{ i) } I_n = \int_1^2 (\log x)^n dx = \int_1^2 (\log x)^n \cdot \frac{1}{x} dx$$

$$\begin{aligned} \text{Let } u &= (\log x)^n, \quad dv = \frac{1}{x} dx \\ \Rightarrow du &= \frac{n(\log x)^{n-1}}{x} dx, \quad v = \log x \end{aligned} \quad \begin{aligned} &\xrightarrow{(1)} \int_1^2 (\log x)^n \cdot \frac{1}{x} dx \\ &= x(\log x)^n \Big|_1^2 - n \int_1^2 (\log x)^{n-1} \cdot \frac{1}{x} dx \\ &= 2(\log 2)^n - n I_{n-1} \end{aligned} \quad (1)$$

$$(ii) \text{ Now } I_4 = \int_1^2 (\log x)^4 dx \quad (1)$$

$$\begin{aligned} \text{by (i) } \Rightarrow I_4 &= 2(\log 2)^4 - 4I_3 \Rightarrow I_1 = 2\log 2 - 1 \\ I_3 &= 2(\log 2)^3 - 3I_2 \quad \therefore I_2 = 2(\log 2)^2 - 4\log 2 + 2 \\ I_2 &= 2(\log 2)^2 - 2I_1 \quad \therefore I_3 = 2(\log 2)^3 - 6(\log 2)^2 + 12\log 2 - 6 \\ I_1 &= 2\log 2 - I_0 \quad \therefore I_4 = 2(\log 2)^4 - 8(\log 2)^3 \\ I_0 &= \int_1^2 \frac{1}{x} dx = x \Big|_1^2 = 1 \end{aligned} \quad \begin{aligned} &+ 24(\log 2)^2 - 48\log 2 + 24 \\ &\hline &1, 1 \end{aligned}$$

(b) (i) $\cos \theta = \frac{3}{x}$ as $PT \perp PO$ a cable is tangent. (1)

(ii) $L = 3(\pi - \theta)$ arc length, as $\cos \theta = \frac{3}{x}$ (1)

$\Rightarrow L = 3[\pi - \cos^{-1}(\frac{3}{x})]$ (1)

(iii) $\frac{dL}{dx} = -3 \cdot \frac{-1}{\sqrt{1 - (\frac{3}{x})^2}} \cdot \frac{-3}{x^2}$ (1)

$= \frac{-9}{\sqrt{x^2 - 9}}$

$= \frac{-9}{x\sqrt{x^2 - 9}}$ as $x > 3$ as $T > R$
 $\Rightarrow \frac{dL}{dx} < 0$ (1)

(iv) $\frac{dL}{dx} < 0 \Rightarrow$ rate of change of L against x is decreasing
 i.e. L decreases as x increases.
 NB: as $x \rightarrow \infty \Rightarrow \theta \rightarrow 90^\circ \Rightarrow L \rightarrow \frac{3\pi}{2}$ (i.e. L is finite $\neq 0$). (1)

(v) $S = L + PT$ $PT = \sqrt{x^2 - 9}$ by pythagoras.

$S = 3\pi - 3\cos^{-1}(\frac{3}{x}) + \sqrt{x^2 - 9}$ (1)

(vi) $\frac{dS}{dx} = \frac{-9}{x\sqrt{x^2 - 9}} + \frac{1}{2}(x^2 - 9)^{-1/2} \cdot 2x$

$= \frac{-9}{x\sqrt{x^2 - 9}} + \frac{x}{\sqrt{x^2 - 9}}$

$= \frac{x^2 - 9}{x\sqrt{x^2 - 9}} = \frac{\sqrt{x^2 - 9}}{x}$ (1)

so $\frac{dS}{dx} = \frac{dS}{dx} \cdot \frac{dx}{dx} = \frac{\sqrt{x^2 - 9}}{x} \cdot 2$ (1)

Question Eight

(a) $ax^4 + 4bx + c = 0$ has a double root at $x = \alpha$

$\Rightarrow 4ax^3 + 4b = 0$ has a single root at $x = \alpha$

$\Rightarrow 4a\alpha^3 + 4b = 0$

$\Rightarrow a\alpha^3 + b = 0 \Rightarrow \alpha^3 = -\frac{b}{a}$ (1)

now $ax^4 + 4bx + c = 0$ (1) (put $x = \alpha$ in eqn)

$\alpha(a\alpha^3 + 4b) + c = 0$

$\alpha(-b + 4b) + c = 0$

$3b\alpha + c = 0$ (1)

$\alpha = -\frac{c}{3b} \Rightarrow \alpha^3 = -\frac{c^3}{27b^3}$

$\therefore -\frac{b}{a} = -\frac{c^3}{27b^3}$ (1)

$\Rightarrow -27b^4 = -ac^3$

i.e. $ac^3 = 27b^4$ as required.

b) $P(x) = (x-a)(x-b)Q(x) + R(x)$

as divisor is quadratic $\Rightarrow R(x) = mx + n$

$\therefore P(x) = (x-a)(x-b)Q(x) + (mx + n)$ (1)

now $P(a) = ma + n$ (1)

$P(b) = mb + n$ (2)

(1) - (2) $\Rightarrow m = \frac{P(a) - P(b)}{a - b}$ (1)

$\therefore n = P(a) - ma$ (1)

$= P(a) - \frac{P(a) - P(b)}{a - b} \cdot a$

Q8 (b) cont

so no $\frac{aP(b) - bP(a)}{a-b}$ (1)

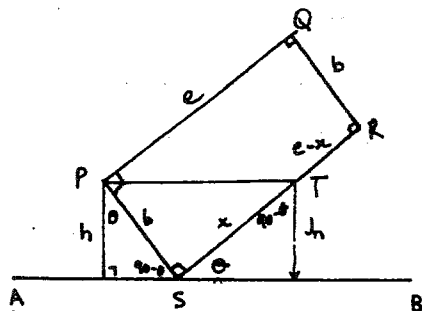
$\therefore R(x) = \left[\frac{P(a) - P(b)}{a-b} \right] x + \frac{aP(b) - bP(a)}{a-b}$ 4

(c) (b) $\sin 3x = -\cos 2x$
 $\sin 3x = -\sin\left(\frac{\pi}{2} - 2x\right)$ by hint
 $= \sin(2x - \pi/2)$ (1)

$\therefore 3x = (2x - \pi/2) + 2n\pi$ or $3x = \pi - (2x - \pi/2) + 2n\pi$
 $\Rightarrow x = 2n\pi - \frac{\pi}{2}$ - (1) $5x = 2n\pi + \frac{3\pi}{2}$
 $x = \frac{2n\pi}{5} + \frac{3\pi}{10}$ - (2)

(ii) smallest soln when $n=0$ in (2) $\Rightarrow x = \frac{3\pi}{10}$ (1)

(d)



Q8 (d) cont

$\sin(90-\theta) = \frac{h}{b} \Rightarrow \cos \theta = \frac{h}{b}$

$\therefore h = b \cos \theta$ (not needed)

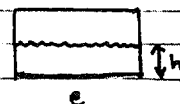
also $\cot \theta = \frac{x}{b}$

$x = b \cot \theta$ (1)

Area $\Delta SPR = \frac{bx}{2} = \frac{b \cdot b \cot \theta}{2} = \frac{b^2 \cot \theta}{2}$

(1)

lie tank flat



Area of Water must be the same as $\left. \begin{aligned} h e &= \frac{b^2 \cot \theta}{2} \end{aligned} \right\}$ (1)

$\therefore h = \frac{b^2 \cot \theta}{2e}$ as required.

#

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